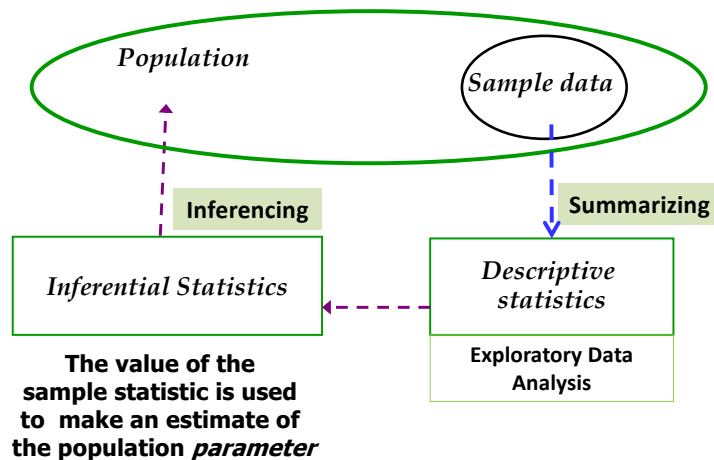


Sampling Distributions and Estimations

MGMT 670: Business Analytics



Scheme for Statistical Analysis



Topics

- Sampling methods
- Central Limit Theorem (CLT)
- Sample statistics and their sampling distributions
- Interval estimates: Confidence intervals (CI)
- Sample size determination

Real Scenario: auto-parts

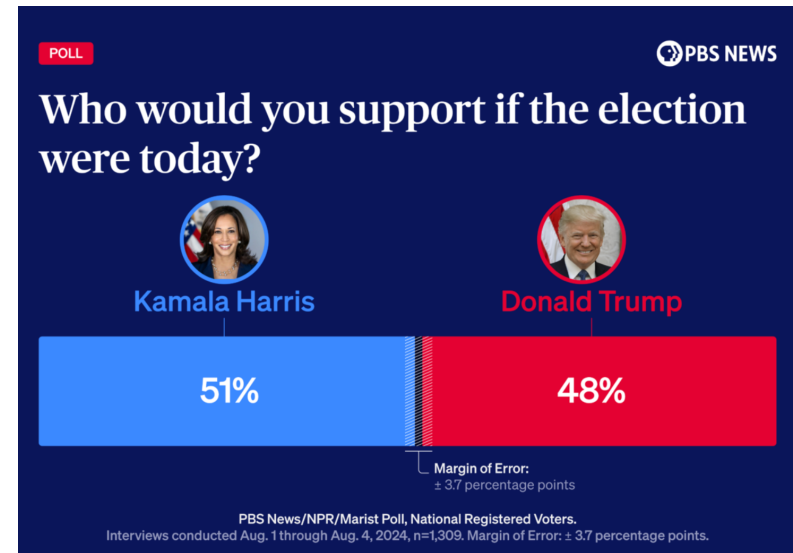
Simple sampling:

- Select random $n=45$ auto-parts stores in Indiana and use sample mean to estimate the average sales.

$$\bar{X} = 2.5 \text{ millions}$$

Real Scenario: political campaign

- Political Science, Inc. (PSI) specializes in voter polls and surveys designed to keep political office seekers informed of their position in a race.
- In a recent election campaign, PSI found that 220 registered voters, out of 500 contacted, favored a particular candidate (A) instead of the other.
- $\bar{p} = 220/500 = .44$ is a point-estimate of the real fraction of people voting for A.



<https://projects.fivethirtyeight.com/polls/president-general/2024/national/>

Point Estimation

- Estimate the population parameter by a single value (statistic), based on a sample data.
- Examples
 $\bar{X} = 2.5mil$ is a point estimate of the population mean μ
 $\bar{p} = .44$ is a point estimate of the population proportion p
- Weakness: Gives no information about how close the estimate is to the unknown population parameter.** ☹️☹️☹️☹️☹️☹️☹️☹️

BIG IDEA

- Think of $\bar{X} = 2.5mil$ or $\bar{p} = .44$ as an instance coming from all possible point estimations, **how likely** is it that they are not very far from the real mean and real proportion?

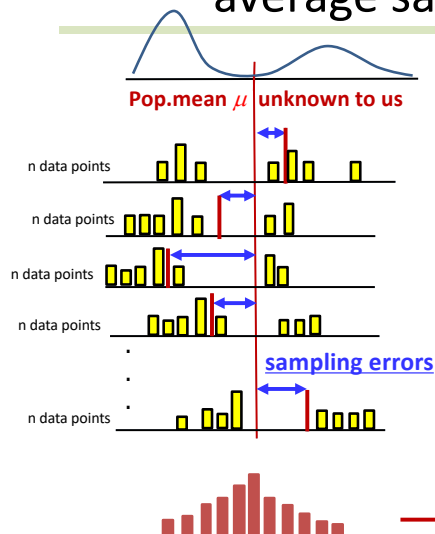
Sampl^{ing} distribution

- **NOT the distribution of one particular sample data**
- It's an distribution, that you would get if you keep
 - sampling, calculating a statistic;
 - sampling, calculating a statistic;
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- sampling, calculating a statistic;
- ...



Example: estimating auto-parts average sales using $\bar{X}$:



If we have information about
sampling distribution of $\bar{X}$,
we can measure:

*The probability that $\bar{X}$
is close to the real average*

We can calculate the probability
that sampling errors is at most
a certain number

→ Sampling distribution of $\bar{X}$

BUT.....

- Be careful not to confuse between:
 - Distribution of a sample data
 - Sampl^{ing} distribution
- If we have to do so much sampling to know
the sampling distribution we can just simply
sample the whole population ☹️☹️☹️☹️

Central Limit Theorem: Sampling distribution of $\bar{X}$

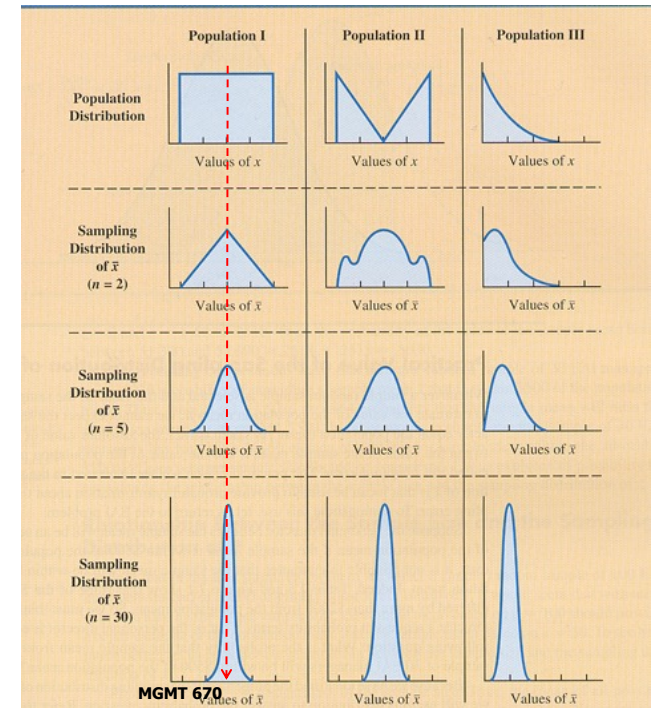
When n becomes large ($n \geq 30$), the sampling distribution of the sample mean $\bar{X}$ can be approximated by a normal distribution, regardless of the population distribution!!!

sampling distribution of $\bar{X} \sim N(\mu, \sigma_{\bar{X}})$

μ = population mean

$$\sigma_{\bar{X}} = \frac{\text{population stdev}}{\sqrt{n}} = \frac{\sigma}{\sqrt{n}} \quad (\text{called standard error of } \bar{X})$$

Same conclusions
for other population
distributions
Fig 7.3 in SBE



In Practice

Population Distribution	Sampling Distribution of Sample Mean
Normal	Exactly Normal (any n)
Non-Normal	Approximately Normal (when $n > 30$)
Non-Normal	? (when $n < 30$)

If sample size is small (<30) and we do not know how the population distribution looks like, then sorry there is no answer. Get more samples!!!

Hospital Stay



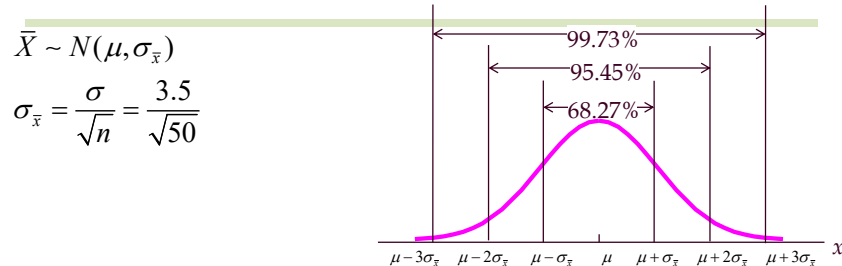
The average length of stay in general community hospitals in the United States is **5.9 days** and the standard deviation is **3.5 days**.

A simple random sample of 50 patients is taken. What is the probability that the **average length of stay of this group of patients** will not exceed 6.7 days?

Hospital Stay

- What is the distribution of $\bar{X}$?
- Why?
- What are the parameters of the distribution?
- What is $P(\bar{X} \leq 6.7)$?

Interval estimation: trying to catch μ



Probability of catching μ using an **1 sample standard deviation** radius “net”.

$$\text{prob that } \mu \text{ is in } \left[\bar{X} - \frac{3.5}{\sqrt{50}}, \bar{X} + \frac{3.5}{\sqrt{50}} \right] = ?$$

68.27% confidence interval of μ is: $\left[\bar{X} - \frac{3.5}{\sqrt{50}}, \bar{X} + \frac{3.5}{\sqrt{50}} \right] = [7 - 0.49, 7 + 0.49]$



Hospital Stay: a more realistic scenario

The average length of stay in general community hospitals in the United States is ~~5.9 days~~ **UNKNOWN** and the standard deviation is **3.5 days**.

A simple random sample of 50 patients is taken, the **average length of stay of this group of patients is 7 days**

What are the possible values of μ (the real average length of stay)?



Hospital Stay: Interpretation

We collect a simple random sample of 50 patients to estimate the average length of stay in general community hospitals in the United States. The sample mean is 7 days, the sample standard deviation is 0.49. The resulting 68.27% Confidence Interval for the average length of stay is $(7 - 0.49, 7 + 0.49)$. Which one is correct?

- 68.27% patients stay in hospital between 6.51 and 7.49 days
- There is 68.27% chance that the true average length of stay is between 6.51 and 7.49
- In repeated sampling, this method produces intervals that capture the true average in about 68.27% of the time
- In repeated sampling, this method produces sample mean between 6.51 and 7.49 in about 68.27% of the time

Interpretation

68.27% confidence interval of μ is: $\left[\bar{X} - \frac{3.5}{\sqrt{50}}, \bar{X} + \frac{3.5}{\sqrt{50}} \right] = [7 - 0.49, 7 + 0.49]$

Wrong interpretation:

~~There is 68.27% chance that true mean is in this interval~~

True mean is fixed, not random!

Correct interpretation:

In repeated sampling, this method produces intervals that capture the true average in about 68.27% of the time.

A baseball coach was curious about the true mean speed of fastball pitches in his league. The coach recorded the speed in kilometers per hour of each fastball in a random sample of 100 pitches and constructed a 95% confidence interval for the mean speed. The resulting interval was (110,120). Which of the following is a correct interpretation of the interval (110,120)?

- If the coach took another sample of 100 pitches, there's a 95% chance the sample mean would be between 110 and 120 km/hr.
- About 95% of pitches in the sample were between 110 and 120 km/hr.
- We're 95% confident that the interval (110,120) captured the true mean pitch speed.

Explanation:

If we were to repeat this sampling and estimation process many times, about 95% of the intervals we construct would contain the true mean.

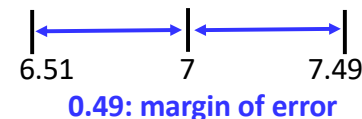
For any single interval, we don't know whether it does or not.

Therefore, with this one estimate, we say that we are 95% confident the interval (110, 120) contains the true mean pitch speed.

Interval Estimation

- Given a point estimation, use an interval to estimate the possible value of the parameter
- Such interval is associated with a "level of confidence"

68.27% confidence interval of μ is: $\left[\bar{X} - \frac{3.5}{\sqrt{50}}, \bar{X} + \frac{3.5}{\sqrt{50}} \right] = [7 - 0.49, 7 + 0.49]$

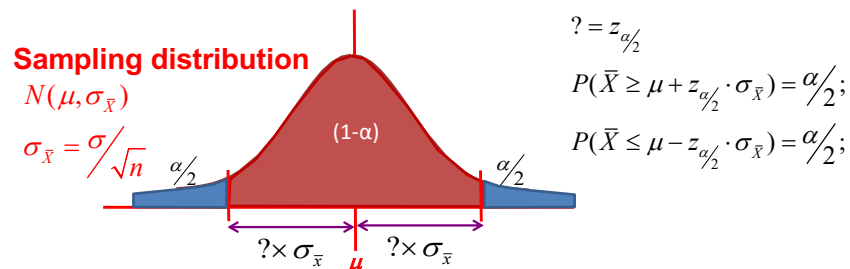


- What's 90% confidence interval for μ ?
- What's 95% confidence interval for μ ?
- What's 99% confidence interval for μ ?

(1-α)×100% confidence interval for mean

α=0.1, we get 90% confidence interval
 α=0.05, we get 95% confidence interval
 α=0.01, we get 99% confidence interval

How large should the net be to get (1-α)×100% confidence interval ? How many standard error to use?



(1-α)×100% confidence interval is $\left[\bar{X} - z_{\alpha/2} \cdot \sigma_{\bar{x}}, \bar{X} + z_{\alpha/2} \cdot \sigma_{\bar{x}} \right]$

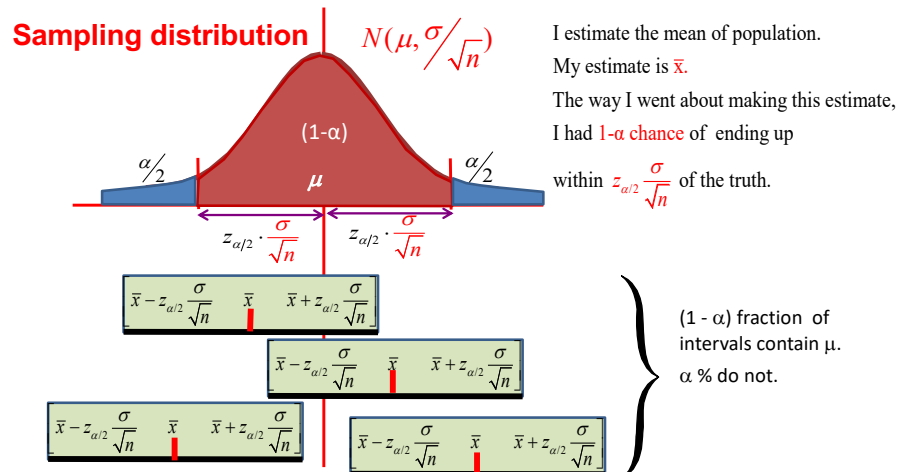
Compute $z_{\alpha/2}$

$z_{\alpha/2}$ = how many standard deviation above the mean
 do I need to go, such that the tail area is $\alpha/2$?

In the table: need to look for the inverse value of $1-\alpha/2$

α	$\alpha/2$	Table Look-up Area $1-\alpha/2$	$z_{\alpha/2}$	Confidence Level (1-α)
.10	.05	.95	1.645	90%
.05	.025	.975	1.96	95%
.01	.005	.995	2.576	99%

Confidence for Population Mean: σ known; large sample (n>30)



- 95% confidence interval for average length of hospital?
- 99% confidence interval for average length of hospital?

Factors Affecting Intervals

$$\left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

Margin of error depends on

- Level of confidence α
- Population dispersion σ
- The precision of our estimate depends on the sample size, but **NOT** on the size of the population being studied.
- **This is WONDERFUL!!!**



Hospital Stay: even more realistic scenario

The average length of stay in general community hospitals in the United States is ~~5.9 days~~ **UNKNOWN** (μ) and the standard deviation is ~~3.5 days~~ **UNKNOWN** (σ)

A simple random sample of 50 patients is taken, the data is
1, 3, 9, 5, 12, 4, 3, ..., 8, 7, 14.

What is 95% confidence interval of μ ?

Sample size determination Example: Hospital Stay

- Suppose we want an estimate of the population mean (of hospital average stay) such that the **sampling error is ½ day or less** with a .95 probability.
- How large a sample size is needed to meet the required precision?

$$\text{margin of error } MoE = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

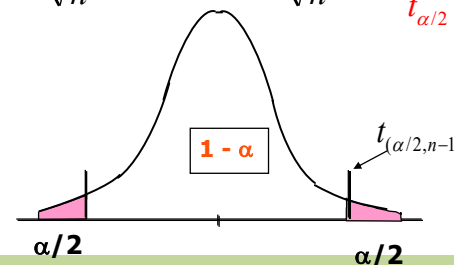
$$n = \left(\frac{z_{\alpha/2} \sigma}{MoE} \right)^2$$

$$n = \left(\frac{\quad \times \quad}{\quad} \right)^2 = ?$$

Interval Estimation of a Population Mean: σ Unknown

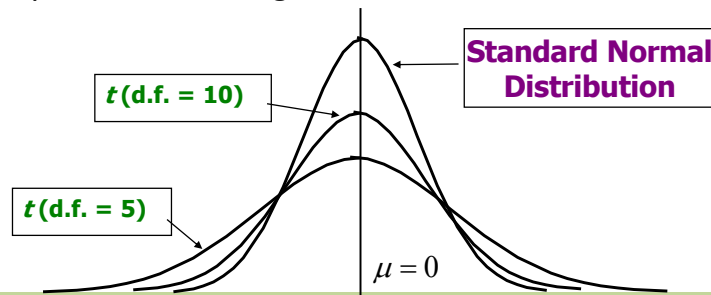
- Use **s** to estimate σ
- Use sample standard deviation and t-distribution
- The interval estimate is given by:

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}} \quad \begin{matrix} t_{\alpha/2} > z_{\alpha/2} \\ t_{\alpha/2} \text{ depends on } n. \end{matrix}$$



t Distribution

- Similar to the standard normal distribution
- Symmetric, Bell-shaped, and Mean = 0
- Has fatter tail than the standard normal distribution
- Depends on the degrees of freedom.



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33

Degrees of Freedom

- Number of observations that are free/independent after sample statistic $\bar{X}$ has been calculated

- Example

Sum of three numbers is 6.

x1 = 1 (or any number)

x2 = 2 (or any number)

x3 = 3 (cannot vary)

Degrees of freedom

$$= n - 1 = 3 - 1 = 2$$



- **Degrees of freedom of t distribution: $n - 1$**

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34



Hospital Stay

The average length of stay in general community hospitals in the United States is ~~5.9 days~~ μ **UNKNOWN** and the standard deviation is ~~3.5 days~~ **UNKNOWN**

A simple random sample of 50 patients is taken, the data is

1, 3, 9, 5, 12, 4,..., 8, 7, 14.

What is the 95% confidence interval of μ ?

$$\left[\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}}; \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}} \right]$$

Sample mean:

Sample standard deviation:

t value:

TABLE 2 t DISTRIBUTION (Continued)

Degrees of Freedom	Area in Upper Tail					
	.20	.10	.05	.025	.01	.005
35	.852	1.306	1.690	2.030	2.438	2.724
36	.852	1.306	1.688	2.028	2.434	2.719
37	.851	1.305	1.687	2.026	2.431	2.715
38	.851	1.304	1.686	2.024	2.429	2.712
39	.851	1.304	1.685	2.023	2.426	2.708
40	.851	1.303	1.684	2.021	2.423	2.704
41	.850	1.303	1.683	2.020	2.421	2.701
42	.850	1.302	1.682	2.018	2.418	2.698
43	.850	1.302	1.681	2.017	2.416	2.695
44	.850	1.301	1.680	2.015	2.414	2.692
45	.850	1.301	1.679	2.014	2.412	2.690
46	.850	1.300	1.679	2.013	2.410	2.687
47	.849	1.300	1.678	2.012	2.408	2.685
48	.849	1.299	1.677	2.011	2.407	2.682
49	.849	1.299	1.677	2.010	2.405	2.680
50	.849	1.299	1.676	2.009	2.403	2.678
51	.849	1.298	1.675	2.008	2.402	2.676
52	.849	1.298	1.675	2.007	2.400	2.674

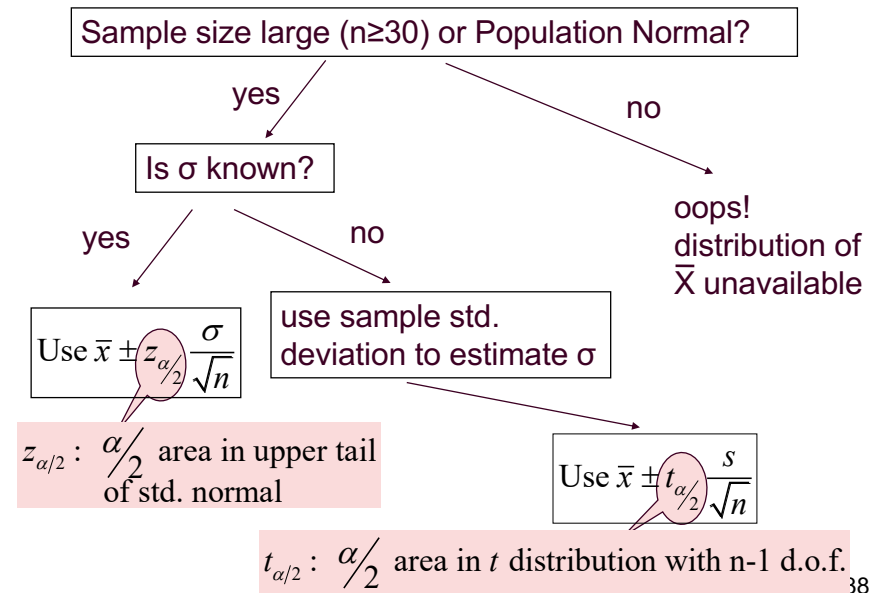
- t Value

- In the t distribution table, we see that $t(0.025, 49) = 2.010$
- In Excel, $T.INV(0.975, 49) = 2.009575$

Topics

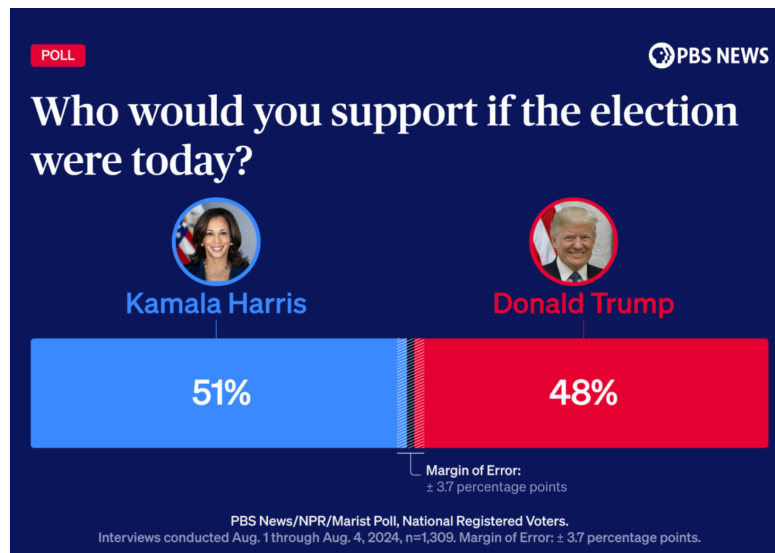
- Sampling distributions
- Interval estimates for mean
 - Known population standard deviation
 - Unknown population standard deviation
- Interval estimates for proportion**
- Sample size determination

Sampling Distributions and Statistical Inference for Population Mean



Example: Political Science, Inc.

- Political Science, Inc. (PSI) specializes in voter polls and surveys designed to keep political office seekers informed of their position in a race.
- In a recent election campaign, PSI found that 220 registered voters, out of 500 contacted, favored a particular candidate instead of the other.
- Develop a 95% confidence interval estimate for the proportion of registered voters that favors the candidate.

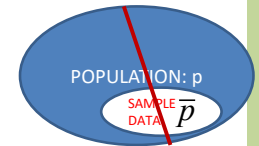


<https://projects.fivethirtyeight.com/polls/president-general/2024/national/>

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41

Proportions



- Categorical variable (e.g., gender)
 - Proportion of population having the property/attribute of interest
- Sample proportion

$$\bar{p} = \frac{\text{number of successes}}{\text{sample size}} = \frac{x}{n}$$

- The sample proportion follows a (scaled) binomial distribution.
 - Possess or don't possess property of interest

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42

Central Limit Theorem: Sampling distribution of $\bar{p}$

When $np \geq 5$ and $n(1-p) \geq 5$, the sampling distribution of the sample proportion $\bar{p}$ can be approximated by a normal distribution.

sampling distribution of $\bar{p} \sim N(p, \frac{\sqrt{p(1-p)}}{\sqrt{n}})$

p = population proportion

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43

Unknown or not unknown? that is the question

~~$(1 - \alpha) \times 100\%$ CI:~~

~~$$\left[\bar{p} - t_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}, \bar{p} + t_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}} \right]$$~~

$(1 - \alpha) \times 100\%$ CI:

$$\left[\bar{p} - z_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}, \bar{p} + z_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}} \right]$$

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44

Example: Political Science, Inc.

- Interval Estimate of a Population Proportion

$$\bar{p} - z_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}} \leq p \leq \bar{p} + z_{\alpha/2} \frac{\sqrt{\bar{p}(1-\bar{p})}}{\sqrt{n}}$$

$$\bar{p} = \frac{220}{500} = 0.44 \Rightarrow n\bar{p} \geq 5, n(1-\bar{p}) \geq 5$$

$$z_{\alpha/2} = 1.96$$

$$\text{Lower Limit} = 0.44 - 1.96 \frac{\sqrt{0.44 \times 0.56}}{\sqrt{500}} = 0.396$$

$$\text{Upper Limit} = 0.44 + 1.96 \frac{\sqrt{0.44 \times 0.56}}{\sqrt{500}} = 0.484$$

So we are 95% confidence that 39.6-48.4% favors this candidate.

Sample Size for an Interval Estimate of a Population Proportion

- Let E = the maximum sampling error

$$\text{We have } E = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

- Solving for n we have

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p(1-p) \rightarrow n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p^*(1-p^*)$$

Note: If no information is available about p^* , then .5 is often assumed because it provides the highest possible sample size.

Example: Political Science, Inc.

- Suppose that PSI would like a .99 probability that the sample proportion is within $\pm .03$ of the population proportion. At 99% confidence, $z_{.005} = 2.576$.
- How large a sample size is needed to meet the required precision?

$$n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p^*(1-p^*)$$

p^*	n
0.1	664
0.2	1180
0.3	1548
0.4	1770
0.5	1843
0.6	1770
0.7	1548
0.8	1180
0.9	664

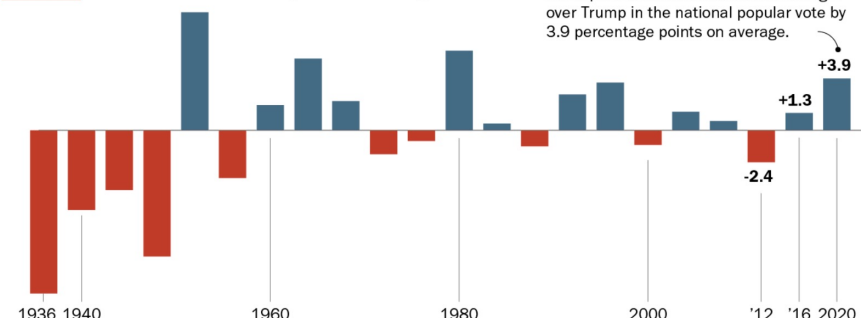
Polling errors in U.S. presidential elections

Bars represent average signed error

Blue bars indicate overestimation of Democratic vote margin.

Red bars indicate overestimation of Republican vote margin.

2020 polls overestimated Biden's margin over Trump in the national popular vote by 3.9 percentage points on average.



<https://www.pewresearch.org/short-reads/2024/08/28/key-things-to-know-about-us-election-polling-in-2024/>

https://www.youtube.com/watch?v=bgHEkOitSXE&ab_channel=CNBC

Summary

- Sampling Distribution
 - Standard error
 - Sampling distribution of sample mean
 - Central limit theorem
- Interval Estimation
 - Sampling error
 - Interval estimation of population mean
 - Interval estimation of population proportion
 - Choosing sample size